

Followup 26-10

Atraumatic periorbital ecchymosis

Patient	48-year-old female. No past medical history of any kind, on no regular medications, no anticoagulants. A week of fatigue, poor appetite, and symptoms she took for a head cold. Itchy eyes one night, which she rubbed a good deal, and bruising around both eyes the following morning.
Outcome	She died. The bruising was the first visible sign of acute promyelocytic leukemia , a subtype of acute myeloid leukemia. Her white count was 237,100 with 86% blasts and Auer rods on the smear, her platelets were 21,000, her fibrinogen was low, and her D-dimer was 35,000. She was in disseminated intravascular coagulation before anyone saw her. She was transferred the same day to a regional cancer center, suffered strokes, pulmonary emboli, a splenic infarct, and intracranial hemorrhage, and died on hospital day 6.
Disposition	Her family met with the intensive care team, said she would not have tolerated the quality of life she was facing, and chose comfort care. She was extubated when they were ready, and she died with them there.

The call

Dispatched priority 2 for a 48-year-old with bruising around her eyes.

She was sitting on her own couch, alert and oriented times four, with a normal airway, breathing, and circulation. She told the crew her eyes had felt itchy two nights earlier and she had rubbed them a great deal, and that she woke the next morning with bruising around both of them. She felt dizzy and was seeing spots in her vision. **She had not hit her head. There was no trauma of any kind.** No drugs, no alcohol. She asked to be checked out at the hospital.

She walked to the ambulance under her own power.

From contact	BP	MAP	Pulse	Respirations	SpO2
+5 min	109/87	94	66	16	not recorded
+13 min	not taken				98%
+14 min	125/89	101	83	not recorded	92%

Transported BLS priority 2. She arrived unchanged and care was transferred at the bedside.

Total patient contact was twenty-two minutes. The record shows an assessment and two sets of vital signs, which is what a walking, talking, oriented patient with no complaint you can treat requires.

Primary impression

Atraumatic periorbital bruising, stable, BLS transport.

That is an accurate description of what was in front of them and there is no version of this call where a crew finds what was wrong with her. **The diagnosis required a blood count.**

It is worth being direct about what this document's purpose is. There is no clinical critique here; no different decision, pickup of an esoteric clinical sign, or prehospital intervention would have changed this case. A 48-year-old with no medical history, who was alert and oriented, walking and talking, breathing comfortably, and saturating in the high nineties, was transported to an emergency department because she asked to be. That was the whole job on this call and it was done.

What the crew could not know is that she was already in disseminated intravascular coagulation. Her platelets were 21,000 against a normal floor of 150,000. Her marrow had been almost entirely replaced by leukemic cells. The bruising around her eyes was not from rubbing them. It was the first place her body showed that it could no longer hold blood inside vessels, and rubbing her eyes was simply the pressure that made it visible there first.

Hospital course

Emergency department, day 0. She was afebrile with normal vital signs on arrival, alert with a GCS of 15, in no distress, and described as non-toxic in appearance but **pale, with purpura scattered diffusely over her body.** The examining clinician documented it carefully: both forearms and wrists, both lower legs and feet, both anterior thighs and knees, the lower chest on both sides, and the lumbar paraspinal region. None of it was tender. Her orbits were not tender and there was no crepitus. Her neurologic examination was normal.

The differential recorded in the note is worth reading, because it is an honest list made before any result came back: mass, thrombocytopenia, lymphoma, leukemia, basilar skull fracture, intracranial hemorrhage, liver failure, or something else. Orders went in for a full laboratory panel, CT head without contrast, ECG, chest radiograph, an antiemetic, and orthostatic vital signs.

The ECG showed sinus rhythm at 84 with **low voltage QRS complexes** and non-specific ST and T wave changes. Her **orthostatic vital signs were positive.** Her INR came back elevated. Her potassium was 2.5.

Then the blood count returned and the laboratory called. **White cell count 237,100**, with a preliminary read of more than 200,000 white cells prompting a request for pathology review. The manual differential found **86% blasts**, an absolute blast count of 203,910, and **Auer rods present.** Platelets 21,000, with the smear estimate reported simply as decreased.

The emergency physician spoke to a hematologist at a regional cancer center and started treatment in the department: hydroxyurea to bring the cell count down, allopurinol, continuous fluids, potassium replacement, and an antiemetic. Ground transport could not collect her for several hours, and with a fibrinogen of 122 and platelets of 21,000 the decision was made to move her by air. **She was told her likely diagnosis in the emergency department and her questions were answered.** The note records that she said her symptoms felt rather mild and that she was mainly troubled by nausea.

Receiving hospital, days 0 to 6. Pathology confirmed **PML::RARA**, the fusion gene that defines acute promyelocytic leukemia. She was started on all-trans retinoic acid, arsenic trioxide, idarubicin, hydroxyurea, and dexamethasone.

She developed a severe headache early on, and a CT found a 1 cm bleed in the right parietal lobe. Neurosurgery elected not to intervene given the size, and recommended keeping her platelets and clotting studies supported. She then became hypoxic, with congestion and infiltrates on chest radiograph.

On day 2 she suddenly became drowsy and could not speak. A stroke code found **occlusions in both middle cerebral arteries and one posterior cerebral artery.** She could not be given a clot-dissolving drug because she was

already bleeding into her head and could not clot. She underwent mechanical thrombectomy of one occlusion with good reperfusion. Imaging then found **pulmonary emboli and a splenic infarct**. With clots forming in that many places at once, and after a discussion with her family about the risks, she was started on heparin despite the bleeding in her brain.

Her breathing failed and she was intubated. Bronchoscopy found blood in her airways. She developed ARDS, then shock requiring three vasopressors, then neutropenia and fever. She was supported with repeated transfusions of platelets, red cells, plasma, and cryoprecipitate.

Overnight on day 5 her left pupil became fixed and dilated. CT found new bleeding, worsening cerebral edema, midline shift, and **herniation**. Her pupils briefly normalized with treatment and then both became fixed and dilated.

Day 6. Her family met with the team and were told the prognosis. They said she would not have tolerated the life she was facing, the loss of independence and the inability to return to work, and they chose comfort care. She was moved to inpatient hospice, her sedation was changed, and **she was extubated when her family was ready**. Her heart stopped shortly afterward and she was pronounced dead that morning.

Cause of death: cardiopulmonary arrest due to cerebral edema, infarction and hemorrhage, and ARDS, all due to acute promyelocytic leukemia.

Education points

Bruising without trauma

Periorbital bruising has a short list of causes and almost all of them are serious.

The reason the eyes show it first is anatomical. The skin of the eyelid is the thinnest on the body, the tissue beneath it is loose, and it has no firm attachment to hold blood in place. Blood that leaks anywhere near there spreads freely and shows through skin thin enough to be translucent. **It takes very little bleeding to produce a dramatic appearance.** That is why raccoon eyes are the classic sign of a basilar skull fracture, where blood tracks forward from the skull base into that loose tissue, and it is also why they are an early sign in patients whose blood is not staying inside vessels anywhere.

When there is no trauma, the differential is essentially systemic:

- **Not enough platelets**, from marrow failure, marrow replacement by cancer, immune destruction, or drugs.
- **Platelets that do not work**, most often from medication.
- **Clotting factors that are missing or being consumed**, from liver failure, anticoagulants, or DIC.
- **Amyloidosis**, where periorbital purpura after trivial pressure is close to a signature finding.
- **Raised venous pressure in the head**, from prolonged coughing, vomiting, or straining.

Notice what is not on that list: rubbing your eyes. In a person with normal platelets and normal clotting, rubbing itchy eyes does not produce bruising. **The rubbing did not cause this. It revealed it.**

This patient had it everywhere the crew and the department looked once somebody looked: forearms, wrists, shins, feet, thighs, knees, lower chest, and lumbar back, none of it tender, none of it explained. **Diffuse, non-tender, atraumatic purpura in a well-looking patient is one of the few findings that tells you something systemic is badly wrong before the patient feels badly wrong.**

Disseminated intravascular coagulation

DIC is the diagnosis that explains why this went the way it did, and it is genuinely counterintuitive, so it is worth building from the beginning.

Normal clotting is local. When a vessel is injured, tissue factor is exposed at that one spot, the cascade fires there, a clot forms there, and the process is switched off by natural anticoagulants so it does not spread. Everything about haemostasis is designed to keep it confined.

DIC is that process losing its boundaries. Something floods the bloodstream with tissue factor or a substance that acts like it, and the cascade fires everywhere at once. Clot forms throughout the small vessels of the body. That alone causes damage, because those microclots block blood supply to organs.

Then the second half happens, and it is the part that catches people out. **Making clot everywhere uses up the materials.** Platelets get consumed into all those clots. Fibrinogen, the protein that clot is actually built from, gets consumed. Clotting factors get consumed. At the same time the body's clot-dissolving system switches on to clear the mess, and it chews through what is left.

The result is a patient who is **clotting and bleeding at the same time, for the same reason.** They are not two problems. They are one process at two stages.

That is exactly this patient's course, and you can read it in order. She bled into her skin before she ever called for an ambulance. She bled into her brain on day 1. Then she clotted, in both middle cerebral arteries, in a posterior cerebral artery, in her lungs, and in her spleen. Then she bled into her airways. Her team started heparin, an anticoagulant, in a woman with active bleeding in her brain, because the clotting had become the more immediate threat. **That decision looks contradictory and it is the correct treatment for DIC,** which is why the underlying cause has to be fixed before anything else will hold.

The laboratory picture is a set of four numbers moving in a pattern, and hers is a textbook example:

- **Platelets fall,** because they are being consumed. Hers were 21,000.
- **Fibrinogen falls,** because it is being turned into clot. Hers was 122, below a floor of 190.
- **Clotting times lengthen,** because factors are consumed. Her prothrombin time was 14.2 seconds and her INR was 1.4.
- **D-dimer rises,** because it is a fragment released when clot is broken down, so it measures how much clot is being formed and destroyed. Hers was **35,000**, against an upper limit of 500. Seventy times the ceiling.

No single one of those makes the diagnosis. The four together, in a patient with a condition known to cause it, is what does.

Why this particular leukemia

Acute promyelocytic leukemia is the reason DIC was present on the day she walked into an emergency department rather than developing later.

In this disease the marrow fills with abnormal cells stalled at one stage of white cell development. Those cells are packed with granules, and the granules contain substances with powerful procoagulant activity. As the cells break down, and especially as treatment kills them, that material spills into the bloodstream and switches the clotting cascade on everywhere. **This is the leukemia that is defined by its bleeding,** and it was first described in patients who bled to death before anyone understood why.

Her smear carried the classic findings: a very high white count, a dominant blast population, and **Auer rods**, which are needle-shaped structures inside the abnormal cells.

Two things about it that a crew should know. The first is that **it is a true emergency in a way most cancers are not**. Most leukaemia presents over weeks and gets worked up over days. This one kills people in the first fortnight, and the most common cause of early death is bleeding.

The second is harder. **Acute promyelocytic leukemia is the most curable form of acute myeloid leukemia**, with the great majority of patients who survive the first two weeks going on to long-term remission on all-trans retinoic acid and arsenic. She was started on both within hours. She died anyway, of the complication that kills people early in this disease, and it had already begun before anybody met her.

Further reading. StatPearls has free chapters on both halves of this case, disseminated intravascular coagulation at <https://www.ncbi.nlm.nih.gov/books/NBK441834/> and acute promyelocytic leukemia at <https://www.ncbi.nlm.nih.gov/books/NBK459352/>

Prehospital considerations

Bruising without trauma is a finding, and it belongs in your report as one. The crew documented that she had not hit her head and had no associated trauma. That single negative is what made the emergency department's list start with thrombocytopenia and leukemia instead of with a skull fracture. Say it out loud at handoff: no mechanism, and the bruising does not match any story.

Look at the rest of the skin. She had purpura on both forearms, both wrists, both shins, both feet, both thighs, both knees, her lower chest, and her lumbar back. None of it hurt, so she did not mention it, and the visible complaint was her face. **A patient who is bruising in one place without cause is bruising in several**, and that is a two-minute look at arms and legs that turns one odd finding into an obvious pattern.

Be careful with the word normal. The narrative records that vitals came back normal. The numbers were 109/87 with a mean arterial pressure of 94, then 125/89 with a saturation that had drifted from 98 to 92. Nothing there is alarming and nobody would act on it differently. But the emergency department found **positive orthostatic vital signs** an hour later on the same patient, which is a real finding that a lying and standing pressure would have shown in the living room. When a patient tells you she is dizzy and seeing spots, the pressure lying down is not the whole answer.

A patient who looks well and whose story does not fit is the one to slow down for. Everything about her presentation was reassuring except the thing she called for, which made no sense. She walked, she talked, she was oriented, her vitals were unremarkable, and she was dying. **The mismatch was the finding.**

The interventions

PREHOSPITAL INTERVENTIONS

Assessment, history, and two sets of vital signs. That is the record, and it is the correct record. There was no intervention available that would have altered anything. The useful work on this call was the history, specifically the absence of trauma and the timeline, and the decision to transport a patient who could have been talked out of going.

Transport. She asked to be checked out and she was taken. Worth naming plainly because the alternative exists on calls like this: a well-appearing patient with a complaint that sounds cosmetic is exactly the patient who gets offered a refusal. She had a platelet count of 21,000 and would have been bleeding into her brain within thirty-six hours.

IN-HOSPITAL INTERVENTIONS

Complete blood count with differential. The test that made the diagnosis, and it cost a tube of blood. When the analyser reported more than 200,000 white cells, the laboratory technologist requested a pathology review rather than releasing it, which is how the blast count and the Auer rods were confirmed.

Orthostatic vital signs. Lying and standing pressures, ordered because she reported dizziness. Positive, and consistent with a patient who was anaemic and volume depleted after a week of poor intake.

CT head without contrast. Ordered to exclude bleeding inside the skull and a basilar skull fracture, both of which were on the differential before the blood count returned.

Hydroxyurea and allopurinol. Hydroxyurea rapidly reduces a dangerously high white cell count. Allopurinol prevents the kidney injury that follows when large numbers of cells break down and release their contents. Both were started in the emergency department rather than waiting for transfer.

Air transfer rather than ground. Ground transport could not collect her for several hours. With a fibrinogen of 122 and platelets of 21,000, and active concern for DIC, the clinician judged that delay unacceptable and arranged a helicopter. **The reasoning is in the note**, which is what makes it a decision rather than a preference.

All-trans retinoic acid and arsenic trioxide. The two drugs that transformed this disease from almost uniformly fatal to usually curable. They work by forcing the stalled abnormal cells to mature rather than by poisoning them.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

The blood count that made the diagnosis

Lab	Value	Reference range
White blood cell (x10 ⁹ /L)	237.1	4.0 to 10.8
Blasts, manual (%)	86	0
Blasts, manual absolute (x10 ⁹ /L)	203.91	0.00
Auer rods	Present	None
Platelet (x10 ⁹ /L)	21	150 to 400
Platelet estimate on smear	Decreased	Adequate
Hemoglobin (g/dL)	11.1	12.0 to 16.0
Hematocrit (%)	33.6	37.0 to 47.0
Nucleated red blood cells (%)	0.1	0.0

A white cell count of 237,100 is not an infection. No infection produces that number. When a white count runs into the hundreds of thousands, the cells being counted are not functioning white cells at all, they are immature ones that the marrow is producing without control, and the count is a measure of how much of the marrow has been taken over.

The blast percentage is the diagnosis. Blasts are the earliest recognisable precursor cells and they should not be in circulating blood in any number. Hers were 86% of a count of 237,100. The absolute number, 203,910, is roughly twenty thousand times the upper limit of zero.

The percentage matters more than the total count, and it is worth understanding why. Marrow works as a production line: a stem cell divides, and its descendants mature through recognisable stages until a finished cell is released into the blood. A blast is the first identifiable step on that line, and under normal conditions it never leaves the marrow.

Finding any in circulating blood means the line has broken down. What defines acute leukemia is not that the marrow is making too many cells, it is that the cells stop maturing and pile up at one stage, so the marrow fills with a single frozen population and has no room left to produce anything else. That is why her platelets were 21,000 and her hemoglobin was 11.1 while her white count was 237,100.

She did not have too many blood cells. She had one kind of useless cell crowding out every other kind, which is why a patient with a white count above two hundred thousand is simultaneously bleeding, anaemic, and unable to fight infection. 20% blasts is the threshold that defines acute leukemia. Hers was 86%.

Two smaller findings on that panel deserve attention. **Auer rods present** are needle-like structures found inside the abnormal cells of acute myeloid leukemia and they are essentially diagnostic. And **nucleated red blood cells** in the circulation, which she had, means the marrow is so disrupted that it is releasing red cells before they have finished maturing, as mature RBCs in mammals do not have nuclei.

Her **hemoglobin of 11.1 is easy to dismiss on its own**. It is only mildly low, and it is low for the same reason her platelets are: the marrow that makes red cells had been crowded out. In a patient bleeding into her skin, a hemoglobin that is nearly normal is not reassurance. It is a starting point.

The clotting picture

Lab	Value	Reference range
Prothrombin time (seconds)	14.2	9.5 to 12.0
INR	1.4	0.9 to 1.2
Fibrinogen (mg/dL)	122	190 to 450
D-dimer (ng/mL FEU)	35,000	0 to 500
Lactate dehydrogenase (U/L)	910	135 to 214
Uric acid (mg/dL)	7.3	2.4 to 5.7

These four numbers together are DIC, and individually none of them would be.

An INR of 1.4 is mild. A fibrinogen of 122 is modestly low. Neither would raise an eyebrow in isolation. But **fibrinogen is an acute phase reactant, which means it should go up when a patient is acutely unwell**, not down. A low fibrinogen in a sick patient means it is being consumed faster than the liver can replace it.

The **D-dimer of 35,000** is the number that removes all doubt. D-dimer is a fragment released when a formed clot is broken down, so it is a measure of clot turnover. Seventy times the upper limit means an enormous amount of clot was being formed and dissolved continuously throughout her body, at a moment when she was sitting up in an emergency department feeling mildly unwell.

Lactate dehydrogenase at 910 is an enzyme released whenever cells are destroyed in quantity, and in leukemia it reflects the sheer rate of cell turnover. **Uric acid at 7.3** is a breakdown product of the genetic material inside those cells, and it is why allopurinol was started before she was transferred.

The rest of the panel

Lab	Value	Reference range
Potassium (mmol/L)	2.5	3.4 to 5.0
Chloride (mmol/L)	96	98 to 107
Anion gap (mmol/L)	16	5 to 14
Alkaline phosphatase (U/L)	206	35 to 104
Aspartate aminotransferase (U/L)	53	0 to 35
Alanine aminotransferase (U/L)	37	0 to 35
hs troponin T, 2 hour (ng/L)	24	Less than 15
Creatinine (mg/dL)	0.96	0.50 to 1.00

Her potassium of 2.5 was the most immediately dangerous number on this panel and it had nothing to do with her leukemia. A week of poor appetite and intake will do it. At that level the risk is cardiac arrhythmia, which is part of why the emergency department replaced it both orally and intravenously before she travelled.

Her **ECG showed low voltage QRS complexes**, meaning the electrical signals recorded at the skin were smaller than expected. It was listed among her diagnoses as an abnormal ECG and was not pursued further on a day when the blood count had taken over.

The rest is background. The mildly raised liver enzymes and alkaline phosphatase fit an organ being infiltrated by leukemic cells. The troponin of 24 with essentially no change over two hours is a mild elevation in an acutely ill patient and was never a cardiac event. Her kidney function was normal, which in a patient with that cell burden is genuinely fortunate and is what allopurinol and fluids were there to protect.

Learning points

- Bruising without trauma is a systemic finding.** The eyelid has the thinnest skin on the body over loose tissue with nothing to hold blood in place, so it shows first and shows dramatically. In a patient with no trauma, the causes are low platelets, non-functioning platelets, consumed clotting factors, amyloidosis, or raised venous pressure in the head. Rubbing itchy eyes does not bruise a person whose blood is working. It revealed this, it did not cause it.
- Look at the rest of the skin, because the patient will not mention it.** She had non-tender purpura on both forearms, wrists, shins, feet, thighs, knees, lower chest, and lumbar back, and she came in about her face. Painless bruising does not get reported. Two minutes of looking turns a single odd finding into a pattern that is impossible to argue with.
- DIC is clotting and bleeding as one process.** Something switches the cascade on everywhere, microclots block small vessels, and the platelets and fibrinogen get consumed making them, so the patient bleeds because

they have been clotting. Four numbers move together: platelets down, fibrinogen down, clotting times up, D-dimer massively up. Hers were 21,000, 122, an INR of 1.4, and a D-dimer of 35,000.

4. **A normal-looking set of vitals is not the same as a normal patient.** Hers were unremarkable and nobody would have acted differently on them. She was in DIC with 21,000 platelets while those vitals were taken.
5. **When the story does not fit, the mismatch is the finding.** Everything about this woman was reassuring except the reason she called, which made no sense on its own terms. She recognised that herself, which is why she asked to be taken. The right call was to take her seriously and transport her, and that is what happened. Not every follow-up ends with something the crew could have done. This one ends with a crew that did the job correctly for a patient nobody could have helped.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>